

COM3023

UNIVERSITY OF EXETER

**FACULTY OF ENVIRONMENT, SCIENCE
AND ECONOMY**

COMPUTER SCIENCE

Examination, May 2023

Machine Learning and AI

Module Leader: Dr Zeyu Fu

Duration: TWO HOURS

Answer ALL questions. No word count specified.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

Candidates are permitted to use *non-programmable* portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

COM3023

Question 1

- (a) What are the two main types of uncertainty in Machine Learning? Where does the uncertainty come from?

(4 marks)

- (b) What is the main difference between Bayesian and Frequentist perspectives in probabilistic inference? What are three key terms needed in Bayesian inference?

(4 marks)

- (c) Denote y be the number of successes from an experiments consisting of n independent trials. The probability for getting a success is θ , and thus y follows a binomial distribution: $p(y|\theta) = \binom{n}{y}\theta^y(1 - \theta)^{n-y}$.

- (i) Find the parameter θ using maximum likelihood estimation.

(4 marks)

- (ii) Assign a Beta distribution for θ : $p(\theta) = \frac{\theta^{\alpha-1}(1 - \theta)^{\beta-1}}{B(\alpha, \beta)}$. The mean for a Beta distribution is $\frac{\alpha}{\alpha + \beta}$. Show that the Beta distribution on θ is the conjugate prior.

(4 marks)

- (iii) Imagine you have 70 successes from 100 independent trials, What is the mean of the posterior distribution, for $\alpha = 5, \beta = 5$?

(4 marks)

(Total 20 marks)

COM3023

Question 2

- (a) What is the main use of link function? What is the main use of Bayesian information criterion?

(4 marks)

- (b) Is Naive Bayes classifier a discriminative model or a generative model? Explain the answer. State one advantage of Naive Bayes.

(4 marks)

- (c) Can we use exact Bayesian inference for logistic regression and linear regression? Explain the answer.

(4 marks)

- (d) The conditional distribution of a variable y is given as: $p(y|\lambda) \propto \exp(-\lambda)\lambda^y$, and the prior distribution for λ is $p(\lambda) = \frac{1}{\lambda}$. Find the mean and variance of the posterior $p(\lambda|y)$ using the Laplace approximation.

(10 marks)

(Total 22 marks)

Question 3

- (a) A hidden Markov model can be represented as: $\lambda = (S, O, A, B, \pi)$, where $S = (s_1, \dots, s_n)$ are the states, $O = (o_1, \dots, o_t)$ are the observations, A is the transition probability matrix, B is the emission probability matrix and π is the initial probability (or start) vector. For the following information given about the hidden Markov model, draw a transition diagram.

- $S = (s_1, s_2)$
- $O = (o_1, o_2, o_3)$
- $A = \begin{bmatrix} 0.2 & 0.8 \\ 0.3 & 0.7 \end{bmatrix}$
- $B = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$
- $\pi = [0.5, 0.5]$

(8 marks)

- (b) For the observation sequence $(o_1 \rightarrow o_2 \rightarrow o_3)$, use the forward algorithm to estimate the likelihood of the hidden Markov model in the previous question. Show your calculations in different steps of the forward algorithm. You can show the answers up to four decimal places.

(14 marks)**(Total 22 marks)**

Question 4

- (a) What is the main difference between Reinforcement learning and supervised/unsupervised learning? How does Q-learning generate data?

(4 marks)

- (b) Assume a system with four actions (a_1, a_2, a_3, a_4) and four states (s_1, s_2, s_3, s_4) . The rewards and state transitions corresponding to actions at different time steps are provided in Table 1.

Time step	Current state	Reward	Action	Transition state
1	s_3	+5	a_1	$s_3 \rightarrow s_2$
2	s_2	-5	a_2	$s_2 \rightarrow s_3$
3	s_3	+8	a_1	$s_3 \rightarrow s_2$
4	s_2	-10	a_3	$s_2 \rightarrow s_1$
5	s_1	+3	a_4	$s_1 \rightarrow s_2$
6	s_2	-8	a_3	$s_2 \rightarrow s_1$

Table 1: Actions, rewards and transition states.

Apply Q-learning to get the Q-values with learning rate of 0.6 and discount factor of 0.4 for each step. You can initialise the Q-table entries to zero. (2 marks for each correct answer).

(12 marks)

- (c) What is the aim of using the epsilon greedy algorithm?

(2 marks)**(Total 18 marks)**

Question 5

(a) What do the following certainty factor (CF) values mean?

(8 marks)

(i) $CF = 0.2$

(ii) $CF = 1$

(iii) $CF = -0.5$

(iv) $CF = 0$

(b) What is a fuzzy singleton? Provide a short explanation. What is the main difference between Sugeno and Mamdani fuzzy inference?

(4 marks)

(c) What's the fuzzy if-then rule in a Sugeno fuzzy model and explain the terms.

(6 marks)

(Total 18 marks)